

CHEM 404: GENERAL CHEMISTRY
Spring 2019

Instructor: Dr. Samantha Reynolds

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Office: Room 580

Office Hours: Mondays/Wednesdays 10am-12pm (no appointment needed, just drop in). Please feel free to contact me by email with questions or to set up an appointment time if you cannot meet during office hours.

Lecture: Tuesdays 1-4pm

Lab, Two Sections: Wednesdays 6-9pm; Thursdays 6-9pm

Lab Instructor: John Welsh

Required Materials:

Lecture Text – Chemistry: Atoms First. OpenStax. ISBN-13: 978-1-938168-15-4.

This is an open access textbook, [available online](#). You may access chapters directly online, download a pdf of the text for free, or order a hard copy. We will be using this text for both semesters. You will also need a scientific calculator for use in class and on tests and exams.

Course Objectives:

The objective for this course is to teach the fundamental concepts and physical laws of chemistry. At the end of the second semester, a student will be able to describe properties of solids, liquids and solutions relating to intermolecular forces. A student will be able to describe and calculate reaction rates and chemical equilibria, classify acids and bases, perform calculations involving free energy and entropy, and write balanced redox reactions. A student will also become proficient in using SI units and showing work for calculations.

Grades:

LECTURE:

The lecture grade is based on a midterm and final exam, worth 200 points each, weekly online homework sets worth 20 points each, and in-class “team tests” worth 50 points each. While the midterm and final will primarily focus on new material covered in that section, please recognize that chemistry is a highly cumulative subject, and many principles and topics throughout the entire year will build on previously learned material. The lecture is worth 80% of your overall CHEM 404 grade.

Lecture Points Overview

Weekly Homework

10 x 20 points	200 points
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Team Tests

4 x 50 points	200 points
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Midterm	200 points
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Final Exam	200 points
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TOTAL	800 points
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The best way to understand the concepts in this course is to practice them. Homework sets will be assigned most weeks to be completed online before the next class meeting. You are welcome to freely collaborate on these homework sets with your peers, course tutors, or by using other resources. Team tests will be completed during class while working together in your assigned teams. A portion of these assignment grades is based on participation.

Additional non-graded assignments in the lecture include reading assigned chapters and working practice problems in preparation for each class meeting. I expect that you are coming to each lecture having familiarized yourself with the course material and having begun to work on that lecture's problem set.

An extra credit assignment will be available that will be worth 25 points. Details of the assignment will be announced early in the semester.

LABORATORY:

The CHEM 404 General Chemistry laboratory is a supplement to the lecture. As such, it counts as 20% of your total grade in the course. Please note that you must pass the lab in order to pass the course. A syllabus for the laboratory will be distributed separately by your lab instructor.

ASSIGNMENT OF A LETTER GRADE:

After your lecture and lab grades have been combined, your overall letter grade for the course will be based on the percentage grading structure below.

A = 93.0-100	B- = 80.0-82.9	D+ = 65.0-69.9
A- = 90.0-92.9	C+ = 77.0-79.9	D = 60.0-64.9
B+ = 87.0-89.9	C = 73.0-76.9	F = 59.9 and below
B = 83.0-86.9	C- = 70.0-72.9	

Late Work:

It is imperative that you keep up with assigned homework each week. Homework will be available in the style of an online "quiz" that will be available to submit until the next lecture meeting. At that time, no late submissions will be accepted, and students that have not submitted answers to the homework set will receive a zero grade for that assignment.

Attendance:

Missing even one lecture may put a student at a significant disadvantage in the course. Please plan to attend all lectures. If you are unable to attend class, it is your responsibility to arrange for a classmate to take notes for you.

If circumstances beyond a student's control prevent him or her from taking a team test or exam, special arrangements can be discussed so that the student is not penalized for an excused absence. Please let me know as soon as possible if you will have to miss a lecture due to a legitimate excuse. If circumstances do not allow you to make prior arrangements for your absence, please contact me as soon after the missed class as you are able.

myCourses (Canvas):

myCourses is UNH's course management system. You can access your UNH myCourses page at mycourses.unh.edu and use your UNH ID and password to log in. All course announcements, assignments, and documents, including homework "quizzes", lecture problem sets, and answer keys will be made available on myCourses during the semester. Academic Computing Services can help if you have issues with access to myCourses; it is your responsibility to resolve any technical issues.

Use of Technology in the Classroom:

Electronic note-taking and audio recording are allowed with permission of the instructor. Other electronic devices (cell phones, mp3 players, etc.) are not to be used during class. If you must leave your phone on for an emergency situation, please keep it on silent so as not to disrupt the class.

Academic Honesty:

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<http://www.unh.edu/vpsas/handbook/academic-honesty>). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work – in whole or in part – as your own; collaborating on answers for individual assignments, or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes, the textbook, or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and put away for the duration of the exam. Ear buds are not permitted.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Academic Support/Tutoring:

The Center for Academic Enrichment is a resource for all students enrolled in UNH Manchester courses. Through a collaborative tutoring approach, they aim to provide students with opportunities to maximize their learning potential, creating life-long learners for academic and professional success. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in math and science courses. **All enrolled students are entitled to 1 hour of free one-to-one tutoring per week, per course.** In addition, peer and professional tutors may run study groups for you class and provide drop-in math tutoring. Please visit the CAE online (<http://manchester.unh.edu/current-students/cae-more-powerful-learning>) to learn more about their services or to make an appointment with a tutor.

Student Accessibility Services at UNH Manchester:

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you have a disability and wish to receive accommodations, you must register with the Student Accessibility Services Office. The coordinator at UNHM is Jenessa Zurek, who can be contacted at (603) 641-4383 or jenessa.zurek@unh.edu. Please note that class/exam accommodations cannot be made by a professor without appropriate documentation from this office.

Mental Health Services at UNH Manchester:

In partnership with The Mental Health Center of Greater Manchester (MHCGM), UNH Manchester offers consultation visits with a licensed mental health therapist on a walk-in basis in the Academic Advising department (located in the Student Services suites, 4th floor). Current semester walk-in hours are Tuesdays from 9:00am-5:00pm and Thursdays from 9:00am-1:00pm. You may call ahead to be placed on the schedule by calling Academic Advising at (603) 641-4170.

Class Cancellations and Delayed Openings:

UNH Manchester makes weather-related decisions independent of UNH Durham. For information on cancellations and closings, call the UNH Manchester Information Line at **603 641 4100**, or visit the UNH Manchester website.

Closing and delay information will also be announced on the following stations:

WMUR Channel 9 TV; WGIR 101 FM and 610 AM; WJYY 105.5 FM; WMLL 96.5 FM; WOKQ 97.5 FM; WZID 95.7 FM; WFEA 1370 AM.

TENTATIVE LECTURE SCHEDULE
CHEM 404 – Spring 2019

Date	Topics	Textbook Chapters	Homework
1/22	Introduction; Liquids & Intermolecular Forces	10.1-10.4	
1/29	The Properties of Solutions	11.1-11.4	HW 1 Due
2/5	Team Test I Kinetics: Rates of Chemical Reactions	17.1-17.3	HW 2 Due
2/12	Kinetics (continued)	17.4-17.7	
2/19	Team Test II Chemical Equilibria	13.1-13.3	HW 3 Due
2/26	Chemical Equilibria (continued)	13.4	HW 4 Due
3/5	MIDTERM	10, 11, 13, 17	*HW 5 Due <i>Sunday 3/3</i>
3/12	<i>Spring Break – No Lecture or Lab</i>		
3/19	Acid-Base Equilibria	14.1-14.3	
3/26	Acid-Base Equilibria (continued)	14.6-14.7	HW 6 Due
4/2	Equilibria of Precipitation and Dissolution	15.1	HW 7 Due
4/9	Team Test III Thermodynamics	12.1-12.2	HW 8 Due
4/16	Thermodynamics	12.3-12.4	
4/23	Team Test IV Electrochemistry	16.1	HW 9 Due
4/30	Electrochemistry (continued)	16.2-16.4	HW 10 Due
5/7	FINAL EXAM	12, 14-16	